



Is AI a ‘substance driver’ or a ‘fiction booster’? The impact of AI application on corporate green innovation bubbles

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ABSTRACT

This study investigates whether artificial intelligence (AI) acts as a "substance driver" that promotes genuine green innovation or a "fiction booster" that facilitates sophisticated greenwashing. Analyzing panel data from Chinese A-share listed manufacturing companies spanning 2013 to 2023, fixed effects regression results reveal that AI application significantly suppresses green innovation bubbles, demonstrating a substantial reduction effect at the sample mean. However, this effect is highly contingent on external institutional and competitive contexts. Stringent environmental regulation amplifies AI's bubble-suppressing effect by channeling AI capabilities toward substantive compliance-oriented innovation. Additionally, intense market competition amplifies this effect, as competitive pressure disciplines firms to deploy AI toward substantive innovations that generate genuine competitive advantages rather than superficial green credentials. Heterogeneity analyses reveal AI's effectiveness is significantly stronger among non-state-owned enterprises and non-heavy-polluting firms, while SOEs and heavy-polluting firms show insignificant effects. Firms located in AI innovation pilot zones exhibit stronger bubble-suppressing effects compared to those outside. These findings contribute an integrated "institution-market" contingency framework to technology adoption literature, introduce an objective patent-based measure of innovation quality, and demonstrate that AI's role in corporate sustainability is neither technologically deterministic nor universally beneficial but critically depends on the alignment of institutional pressures and market incentives. The results remain robust to instrumental variable analysis, propensity score matching, Heckman correction, and multiple robustness checks.

1. Introduction

1.1. The paradox of green innovation in the era of carbon neutrality

The global imperative to address climate change has placed corporate green innovation at the forefront of both policy agendas and strategic management discourse (Amore and Bannedsen, 2016; Liu et al., 2024; Tian et al., 2024; Dai et al., 2025a). As nations worldwide commit to ambitious decarbonization targets, such as China's pledge to achieve carbon peak by 2030 and carbon neutrality by 2060, governments have deployed an unprecedented array of policy instruments to stimulate corporate investment in

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environmentally friendly technologies. These instruments include direct subsidies for green R&D, preferential tax treatment for environmental projects, low-interest green loans, and accelerated depreciation for clean technology assets. This policy activism has catalyzed a remarkable surge in corporate green activities, as evidenced by the explosive growth in green patent filings, environmental certification acquisitions, and sustainability-themed corporate disclosures. Between 2013 and 2023, green patent applications in China increased by over 300%, signaling an apparent renaissance in environmental innovation.

However, beneath this wave of green enthusiasm lies a critical paradox that threatens to undermine the very objectives these policies seek to achieve. The generous incentives designed to foster genuine environmental progress have simultaneously created a fertile ground for strategic, symbolic behavior. This has given rise to the phenomenon of the “Green Innovation Bubble,” a sophisticated form of greenwashing where firms prioritize the quantity of environmental claims over their substantive quality (Al-Balushi et al., 2025; Geng et al., 2024). Rather than pursuing radical, high-impact research and development that yields measurable environmental benefits through breakthrough technologies and fundamental process innovations, firms increasingly engage in low-quality innovation activities. These activities include filing for utility model or design patents on trivial modifications, cosmetic product re-designs marketed as “eco-friendly,” and incremental adjustments to existing technologies that generate impressive patent counts but negligible environmental impact (Zhou et al., 2024; Xuan et al., 2025). Such behavior is driven by a strategic calculus: in an environment where policy resources are allocated based on observable green credentials rather than verified environmental outcomes, the optimal strategy may be to maximize visible signals of environmental commitment at minimum cost (Arici and Uysal, 2022; Huang and Ma, 2024; Dai, 2025).

The consequences of this green innovation bubble extend far beyond individual firm behavior. At the aggregate level, it leads to a significant misallocation of scarce capital and human resources away from genuinely transformative environmental technologies toward superficial innovations. It distorts market signals, making it difficult for investors and consumers to distinguish between authentic environmental leaders and skilled pretenders. Most critically, it undermines the effectiveness of national and global climate initiatives by creating an illusion of progress while actual environmental challenges remain unaddressed (Dai and Khan, 2023). The existence and prevalence of such bubbles is not merely theoretical speculation but is increasingly acknowledged in scholarly literature as a critical barrier to sustainable development. Recent empirical studies have documented systematic patterns of low-quality green patenting, particularly in policy-sensitive sectors and among firms facing acute legitimacy pressures. The measurement of this phenomenon is intrinsically linked to its underlying cause: the strategic pursuit of easily quantifiable metrics like patent counts by firms responding to policy incentives necessitates a measurement approach that can distinguish between substantive innovation reflected in high-quality invention patents and superficial patenting behavior manifested in lower-quality utility models and design patents.

1.2. The emergence of artificial intelligence as a transformative force

Concurrently, the corporate landscape is being reshaped by the proliferation of Artificial Intelligence (AI), a general-purpose technology with the potential to fundamentally alter processes of production, management, and innovation (Wang et al., 2025). AI's capacity to analyze vast datasets, identify complex patterns, and automate sophisticated tasks promises to enhance corporate productivity and efficiency on an unprecedented scale. In the context of R&D, AI can accelerate discovery, optimize experimental design, and enable the development of more complex and effective solutions, positioning it as a powerful engine for high-quality innovation (Zheng and Zhang, 2023; Dai et al., 2025b).

Yet, the transformative potential of AI is not without its complexities. Research on technology adoption reveals a “productivity paradox,” often characterized by a “J-curve” effect, where the initial implementation of AI leads to a temporary decline in performance. This dip is attributed to significant adjustment costs, the necessity of redesigning legacy workflows, and the challenges of building complementary organizational capabilities, such as data infrastructure and specialized human capital. This paradox suggests that the impact of AI is neither immediate nor uniform. The path from AI investment to tangible improvements in complex domains like innovation is fraught with friction. Firms might initially apply AI to “low-hanging fruit”, such as optimizing the marketing of “green” products or automating the generation of patent application text, before they master the technology for substantive R&D. This raises the critical question of whether AI, in certain contexts, could inadvertently inflate rather than suppress green innovation bubbles, at least in the short term.

1.3. Research question and contribution

The confluence of proliferating green innovation bubbles and rapidly advancing AI technologies raises a critical question: Under what conditions does AI application act as a “substance driver” that suppresses corporate green innovation bubbles, and when does it risk becoming a “fiction booster” that facilitates them?

This question carries profound implications beyond academic interest. If AI inherently promotes substantive innovation, policies encouraging AI adoption would naturally align with environmental objectives. However, if AI's impact depends on contextual factors, effective policy requires understanding when and how AI serves sustainability goals versus undermining them.

To address this question, we develop an integrated theoretical framework synthesizing multiple perspectives. We begin with the Resource-Based View (RBV) and Signaling Theory to establish our baseline prediction. From the RBV perspective, AI represents a valuable organizational capability that enhances firms' intrinsic capacity for substantive innovation through sophisticated analysis, faster experimentation, and optimized resource allocation. This capability should naturally steer firms toward higher-quality invention patents rather than superficial utility models. From the Signaling Theory perspective, AI reduces information asymmetry by enabling stakeholders to scrutinize corporate environmental claims more effectively, thereby raising the cost and risk of greenwashing.

Together, these mechanisms suggest AI should suppress green innovation bubbles.

However, we argue this effect is not unconditional but fundamentally shaped by external contexts. We integrate Institutional Theory and Competition Theory into an "institution-market" dual moderation framework, recognizing that firms simultaneously navigate coercive institutional pressures and competitive market forces. Stringent environmental regulation acts as a channeling mechanism that compels firms to deploy AI for genuine problem-solving to avoid penalties, thereby amplifying its bubble-suppressing effect. Similarly, intense market competition serves as a disciplining mechanism that compels firms to deploy AI toward substantive innovations generating genuine competitive advantages, thereby further amplifying AI's bubble-suppressing effect.

This study makes three primary contributions to the literature.

First, revealing AI's double-edged nature in corporate sustainability. By moving beyond narratives that AI universally promotes innovation quality, we demonstrate that AI can facilitate either substantive progress or sophisticated greenwashing depending on contextual conditions. This introduces critical ethical and strategic dimensions to understanding AI's role in sustainability and challenges technologically deterministic views prevalent in existing literature.

Second, innovating an "institution-market" dual contingency framework. Unlike prior research examining regulation or competition in isolation, we simultaneously consider these distinct yet complementary forces. This integrated approach reveals how institutional pressures and market dynamics jointly shape technology deployment strategies, providing a more comprehensive model of contextual effects. Our framework demonstrates that the same technology can produce opposing outcomes depending on whether institutional and market incentives align or conflict.

Third, adopting an objective, patent-based measure of green innovation bubbles. Following Geng et al. (2024), we operationalize innovation bubbles as the standardized difference between green patent quantity and quality, where higher values indicate that a firm's patent counts substantially exceed its substantive innovation quality. This measurement approach captures the essence of strategic patenting behavior where firms inflate easily quantifiable metrics without corresponding improvements in innovation substance, while relying on objective, publicly available patent data rather than subjective survey measures.

These contributions advance scholarly understanding of technology adoption, corporate sustainability, and innovation quality while providing actionable insights for policymakers seeking to harness AI for genuine environmental progress.

2. Literature review and hypothesis development

2.1. The duality of green innovation: substantive progress versus strategic bubbles

Green innovation is broadly defined as the development of new or modified products, processes, technologies, and organizational practices that reduce environmental burdens and contribute to ecological sustainability. It is a cornerstone of corporate strategy for achieving a "win-win" of enhanced environmental performance and sustained competitive advantage. Substantive green innovation involves fundamental technological advancements that lead to measurable improvements in resource efficiency, pollution reduction, and energy conservation (Lin and Zhu, 2025).

However, the institutional and market rewards associated with a "green" image have created powerful incentives for "greenwashing", the selective disclosure of positive environmental information without corresponding substantive action, aimed at misleading stakeholders. Greenwashing manifests in various forms, from vague "eco-friendly" labeling to emphasizing minor positive attributes while hiding significant negative impacts. The green innovation bubble (GIB) can be conceptualized as a specific and sophisticated form of greenwashing within the R&D domain. It is driven by information asymmetry between the firm and its stakeholders (investors, regulators, consumers), where the firm possesses private information about the true quality and impact of its innovations (Liang et al., 2024). To secure legitimacy and policy support, firms may strategically inflate their green patent portfolios with low-cost, low-quality patents that are easy to obtain but offer little to no real environmental benefit, thus creating a "bubble" of perceived innovation (Sun et al., 2024).

2.2. Artificial intelligence and corporate innovation: a resource and information perspective

The application of AI is revolutionizing corporate R&D. AI technologies can process and analyze massive datasets, run complex simulations, and identify novel solutions far beyond human capacity, thereby accelerating the pace and improving the quality of innovation. The impact of AI on innovation can be understood through two complementary theoretical lenses: the Resource-Based View (RBV) and Signaling Theory.

From the perspective of the **Resource-Based View (RBV)**, sustainable competitive advantage stems from a firm's control over valuable, rare, inimitable, and non-substitutable (VRIN) resources and capabilities. AI, particularly when deeply integrated into a firm's proprietary datasets and workflows, represents such a capability. It enables a firm to more effectively and efficiently transform its inputs (e.g., R&D spending, human capital) into high-quality innovation outputs. By optimizing resource allocation and enhancing analytical power, AI strengthens a firm's core competence in creating substantive technological advancements (Tian et al., 2023).

From the perspective of **Signaling Theory**, firms operate in an environment of information asymmetry, where they use signals (e.g., disclosures, certifications, patent filings) to convey unobservable qualities to external stakeholders. Greenwashing and GIBs represent a form of "false signaling," where the signal (a large number of green patents) does not accurately reflect the underlying quality (substantive environmental commitment). A key function of AI is its ability to process vast amounts of structured and unstructured data, thereby increasing transparency and reducing information asymmetry (Begum et al., 2022). AI-powered analytics can enable stakeholders to more effectively scrutinize corporate claims, compare "green talk" with "green walk" (e.g., by analyzing

satellite imagery of emissions or supply chain data), and detect inconsistencies. This increases the cost and risk associated with false signaling, as the probability of detection and subsequent reputational damage rises.

2.3. The main effect: how AI application suppresses green innovation bubbles (H1)

Integrating the RBV and Signaling Theory perspectives provides a robust theoretical foundation for predicting the main effect of AI on GIB. The logical chain proceeds as follows: First, from an RBV standpoint, AI functions as a powerful internal capability that directs R&D efforts toward higher-quality outcomes. By enabling more sophisticated analysis and simulation, AI enhances a firm's ability to tackle complex technological challenges, leading to a higher propensity for generating substantive *invention* patents rather than superficial utility models or designs. AI is a resource that inherently improves the *substance* of the innovation process. Second, from a Signaling Theory standpoint, AI acts as an external and internal verification mechanism. Internally, it provides managers with more accurate data on their firm's environmental footprint, enabling better decision-making. Externally, it empowers stakeholders to more effectively audit corporate environmental claims. This dual-pronged increase in transparency raises the expected cost of engaging in deceptive, low-quality innovation. By making greenwashing more difficult and costly, AI disincentivizes the very behavior that creates innovation bubbles. Together, these mechanisms suggest that AI application should steer firms away from strategic, low-quality green patenting and toward more substantive innovation.

Hypothesis 1 (H1): *Ceteris paribus, AI application is negatively associated with the corporate green innovation bubble.*

2.4. The contingent nature of technological impact: an 'institution-market' framework

While AI may have an innate tendency to suppress GIBs, the impact of such a powerful technology is rarely deterministic. Its effects are shaped and channeled by the external context in which the firm operates. We propose an "institution-market" framework to theorize how two key contextual forces, the coercive push from government regulation and the competitive pull from the market, moderate AI's impact on corporate innovation strategy.

2.4.1. The coercive push of environmental regulation: a catalyst for substantive innovation (H2)

Institutional Theory posits that organizations conform to the rules, norms, and beliefs of their external environment to gain legitimacy, stability, and resources. Environmental regulations are a powerful form of "coercive isomorphism," where state authority and the threat of sanctions compel firms to adopt specific practices regardless of their immediate efficiency benefits. Stringent environmental laws, such as China's New Environmental Protection Law (NEPL), increase the tangible costs of non-compliance through fines, production halts, and legal liabilities, thereby creating a powerful incentive for firms to achieve genuine improvements in environmental performance (Liu et al., 2021; Liu and Li, 2022).

This coercive pressure directly influences how firms deploy their technological capabilities. In a lax regulatory environment, a firm might use its AI resources for symbolic purposes that enhance its green image at minimal cost. However, when faced with strict, well-enforced regulations, the strategic calculus shifts. The primary goal becomes substantive compliance to avoid costly penalties. In this context, firms are motivated to direct their powerful AI capabilities toward solving real-world environmental challenges, such as optimizing energy consumption in production processes, developing cleaner materials, or creating more effective pollution abatement technologies. These activities are far more likely to result in high-quality, substantive invention patents. Therefore, strong environmental regulation acts as a channeling mechanism, directing AI's potential away from superficial applications and toward substantive innovation, thereby amplifying its inherent bubble-suppressing effect.

Hypothesis 2 (H2): *The negative relationship between AI application and the corporate green innovation bubble is stronger in regions with higher environmental regulation intensity.*

2.4.2. The disciplining force of market competition: a catalyst for efficient AI deployment (H3)

Market competition serves as a powerful external governance mechanism that disciplines corporate behavior and resource allocation decisions. From an economic efficiency perspective, competitive pressure compels firms to deploy their resources, including advanced technologies, toward activities that generate genuine value rather than superficial signals. This disciplining effect operates through multiple channels. First, intense competition heightens market scrutiny and reduces tolerance for strategic deception. In highly competitive markets, customers, investors, and other stakeholders have abundant alternatives and strong incentives to evaluate firms' claims critically. Greenwashing and superficial innovation are more likely to be detected and punished through customer defection, investor skepticism, and reputational damage. This elevated detection risk discourages firms from deploying AI for cosmetic green initiatives and instead channels AI capabilities toward substantive innovations that can withstand market scrutiny. Second, competition creates survival pressure that demands efficient resource utilization. When competitive intensity is high, firms cannot afford to waste valuable AI resources on activities that fail to generate real competitive advantages. Superficial green patents may temporarily enhance corporate image but provide no durable differentiation in competitive markets where rivals can easily imitate or expose such tactics. Consequently, firms facing intense competition are more likely to deploy AI for genuine R&D breakthroughs that create sustainable competitive advantages through superior environmental technologies. Third, competitive markets facilitate learning and benchmarking. Firms in competitive industries can more readily observe and compare innovation outcomes across competitors, making it easier to distinguish between substantive and superficial green innovations. This transparency further raises the costs of bubble-creating behavior and incentivizes genuine innovation efforts. Conversely, in less competitive, highly concentrated markets, firms face weaker external discipline. Market power provides a buffer against scrutiny, allowing firms to engage in symbolic green

activities without immediate consequences. The absence of competitive pressure reduces the urgency to maximize AI's potential for substantive innovation, potentially leading to complacency and superficial deployment of AI capabilities.

Hypothesis 3 (H3): *The negative relationship between AI application and the corporate green innovation bubble is stronger in industries with a higher degree of market competition.*

3. Research design

3.1. Sample selection and data sources

The initial sample for this study comprises all Chinese A-share listed manufacturing companies from 2013 to 2023. Manufacturing firms are selected due to their significant environmental impact and their central role in green innovation activities. The sample period is chosen to capture the period following the rise of AI as a major corporate investment area and to allow for sufficient time-series variation. After excluding firms with missing data for key variables (e.g., AI investment, patent data, financial controls) and firms with special treatment (ST, *ST), the final unbalanced panel dataset consists of firm-year observations.

Data are collected from several authoritative sources:

The China Stock Market & Accounting Research (CSMAR) database provides firm-level financial data, corporate governance information, and AI patent data for constructing the core independent variable.

The **China Research Data Services (CNRDS)** platform is the source for green patent application data, which is essential for constructing the dependent variable.

The **China Environmental Statistics Yearbook** and various **Prefectural City Statistical Yearbooks** provide data on regional environmental enforcement, used to measure the environmental regulation moderator.

3.2. Measurement of variables

The operationalization of all variables is detailed in [Table 1](#).

3.3. Econometric model specification

To determine the appropriate estimation approach for the panel data regression, we conduct three specification tests: the F-test, the Lagrange Multiplier (LM) test, and the Hausman test. The results are presented in [Table 2](#).

The F-test examines whether fixed effects are preferred over pooled OLS, with the null hypothesis that pooled OLS is more appropriate. The LM test evaluates whether random effects are preferred over pooled OLS, with the null hypothesis favoring pooled OLS. The Hausman test compares fixed effects versus random effects, with the null hypothesis that random effects are more appropriate. A p-value below 0.1 indicates rejection of the null hypothesis.

As shown in [Table 2](#), all three tests yield p-values of 0.0000, strongly rejecting their respective null hypotheses. The F-test rejects pooled OLS in favor of fixed effects, the LM test rejects pooled OLS in favor of random effects, and the Hausman test rejects random effects in favor of fixed effects. Collectively, these results indicate that the panel fixed effects model is the most appropriate specification for our baseline regression analysis.

Next, to test our hypotheses, we employ a panel data fixed-effects regression model. This approach is well-suited for controlling for unobserved time-invariant firm-specific heterogeneity (e.g., corporate culture, managerial talent) that could confound the relationship between AI and GIB. Year fixed effects are also included to account for macroeconomic shocks and trends common to all firms.

Table 1
Variable definitions and data sources.

Variable Category	Variable	Symbol	Measurement Method	Data Source
Dependent	Green Innovation Bubble	GIB	Standardized difference between green patent quantity and quality, following Geng et al. (2024)	CNRDS
Independent	AI Application	AI	Natural logarithm of (1 + amount of AI patents)	CSMAR
Moderating	Environmental Regulation	ER	Environmental protection frequency / Total frequency	Government working report
Moderating	Market Competition	HHI	Industry Herfindahl-Hirschman Index	CSMAR
Control	Firm Size	Size	Natural logarithm of total assets	CSMAR
Control	Firm Age	Age	Natural logarithm of (current year – establishment year + 1)	CSMAR
Control	Financial Leverage	Lev	Total liabilities / Total assets	CSMAR
Control	Profitability	ROA	Net income / Total assets	CSMAR
Control	Ownership Concentration	Own_Conc	Shareholding percentage of the largest shareholder	CSMAR
Control	Board Size	Board	Number of directors on the board	CSMAR
Control	Institutional Investor	INST	The total number of shares held by institutional investors divided by the outstanding shares.	CSMAR
Control	Return on Equity	ROE	(Net Income / Average Shareholders' Equity) × 100%	CSMAR

Table 2
Model specification tests.

Test	Statistic	P-value
F-test	4.88	0.0000
LM test	11705.44	0.0000
Hausman test	112.62	0.0000

The baseline model to test H1 Is specified as follows:

$$GIB_{it} = \beta_0 + \beta_1 AI_{it} + \sum_k \beta_k Controls_{it} + \mu_i + \lambda_t + \epsilon_{it} \quad (1)$$

where GIB_{it} is the green innovation bubble for firm i in year t , AI_{it} is the AI application, $Controls_{it}$ is the vector of control variables, μ_i represents firm fixed effects, λ_t represents year fixed effects, and ϵ_{it} is the error term. H1 predicts that $\beta_1 < 0$. To test the moderating effects for H2 and H3, we introduce interaction terms into the baseline model:

$$GIB_{it} = \beta_0 + \beta_1 AI_{it} + \beta_2 Moderator_{it} + \beta_3 (AI_{it} \times Moderator_{it}) + \sum_k \beta_k Controls_{it} + \mu_i + \lambda_t + \epsilon_{it} \quad (2)$$

For H2, the Moderator is *ER*. We predict $\beta_3 < 0$, indicating that higher regulation strengthens the negative effect of AI on GIB. For H3, the Moderator is *HHI*. We predict $\beta_3 > 0$. Since a higher HHI means less competition, a positive coefficient on the interaction term implies that the bubble-suppressing effect of AI is weaker in less competitive markets, and therefore stronger in more competitive markets.

4. Empirical results and analysis

4.1. Descriptive statistics and preliminary diagnostics

Table 3 presents the descriptive statistics, Pearson correlation matrix, and variance inflation factor (VIF) test results for the key variables in this study. As shown in Panel A, the mean of GIB is -0.078 with a standard deviation of 0.271, indicating considerable variation in green innovation bubble levels across firms. The variable AI has a mean of 0.308 and a median of 0, suggesting that a substantial proportion of firms have not yet adopted AI technologies, while those that have show considerable variation (S.D. = 0.667), reflecting the heterogeneous adoption of AI technologies across firms. Environmental regulation (ER) has a mean of 0.813, while market competition (HHI) averages 0.005 with relatively low dispersion (S.D. = 0.018).

Regarding control variables, the average firm size (Size) is 22.20, with leverage (Lev) averaging 0.409, indicating that sample firms finance approximately 41% of their assets through debt. The mean return on equity (ROE) is 0.057, and institutional ownership (INST) averages 0.427, suggesting that institutional investors hold substantial stakes in sample firms. The largest shareholder (Top1) holds on average 33.6% of shares, reflecting relatively concentrated ownership structures typical of Chinese listed firms.

The correlations between the independent variables are generally low to moderate, suggesting that multicollinearity is not a significant concern in our regression models. Panel B reports the VIF test results to further confirm this assessment. The mean VIF is 1.280, and all individual VIF values are well below the conventional threshold of 10, with the highest being Size at 1.720. These results confirm that multicollinearity does not pose a concern for our empirical analysis.

4.2. Baseline regression results: the impact of AI on green innovation bubbles

Table 4 presents the fixed-effects regression results testing the main effect of AI application on green innovation bubbles. We progressively introduce control variables from Model (1) to Model (5) to examine the stability of our key findings.

The results provide strong support for Hypothesis 1. Across all five specifications, the coefficient on AI is negative and statistically significant, indicating that AI application significantly suppresses green innovation bubbles. In Model (1) without control variables, the coefficient is -0.008 ($p < 0.05$). After progressively introducing firm-level controls, the coefficient stabilizes at -0.010 ($p < 0.01$) in Models (2) through (5), demonstrating the robustness of our main finding. This result suggests that a one-unit increase in AI application is associated with a 0.010 reduction in the green innovation bubble index, confirming that AI acts as a "substance driver" that redirects corporate green innovation toward higher quality outputs.

Among control variables, firm size (Size) exhibits a positive and significant association with GIB, suggesting that larger firms may face greater pressures for visible green credentials, potentially increasing their propensity for strategic patenting. Profitability measured by ROA shows a negative and significant coefficient, indicating that financially healthier firms are less inclined to engage in superficial innovation bubbles. Institutional ownership (INST) is positively associated with GIB, while leverage (Lev) shows a negative but marginally significant relationship. Board size (Board) and ownership concentration (Own_Conc) do not exhibit statistically significant effects.

The R-squared value of approximately 0.502, combined with firm and year fixed effects, indicates that the model explains substantial variation in green innovation bubbles while controlling for time-invariant firm heterogeneity and common time trends.

Table 3

Descriptive statistics, correlation matrix, and VIF test.

Panel A: Descriptive Statistics and Correlation Matrix													
Variable	Mean	S.D.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) GIB	-0.078	0.271	1.00										
(2) AI	0.308	0.667	0.036***	1.00									
(3) ER	0.813	0.232	0.022***	-0.036***	1.00								
(4) HHI	0.005	0.018	-0.032***	-0.025***	0.002	1.00							
(5) Size	22.20	1.258	0.293***	0.083***	0.011*	-0.060***	1.00						
(6) Lev	0.409	0.204	0.143***	-0.004	-0.004	-0.033***	0.490***	1.00					
(7) ROE	0.057	0.136	0.031***	0.055***	0.026***	0.012**	0.079***	-0.255***	1.00				
(8) Board	2.107	0.198	0.063***	0.001	0.042***	-0.010*	0.269***	0.140***	0.028***	1.00			
(9) Cashflow	0.048	0.068	-0.007	0.033***	-0.019***	0.047***	0.083***	-0.163***	0.350***	0.034***	1.00		
(10) INST	0.427	0.247	0.129***	0.008	0.031***	0.011**	0.442***	0.182***	0.134***	0.235***	0.113***	1.00	
(11) Top1	0.336	0.147	0.062***	0.019***	0.028***	0.024***	0.196***	0.041***	0.163***	0.019***	0.111***	0.488***	1.00
Panel B: Variance Inflation Factor (VIF) Test													
Variable	Size	INST	Lev	Own_Conc	ROE	ROA	Board	AI	HHI	ER	Mean		
VIF	1.720	1.620	1.510	1.350	1.260	1.170	1.110	1.010	1.010	1.010	1.280		
1/VIF	0.582	0.616	0.660	0.742	0.792	0.853	0.898	0.986	0.991	0.994			

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. VIF values below 10 indicate no serious multicollinearity concern.

Table 4
Baseline regression results.

VARIABLES	(1) GIB	(2) GIB	(3) GIB	(4) GIB	(5) GIB
AI	-0.008** (-2.467)	-0.010*** (-3.044)	-0.010*** (-3.059)	-0.010*** (-3.022)	-0.010*** (-3.022)
Size		0.054*** (9.084)	0.054*** (9.095)	0.050*** (8.653)	0.050*** (8.653)
Lev		-0.029** (-2.082)	-0.029** (-2.088)	-0.023 (-1.614)	-0.023 (-1.614)
ROE		0.043*** (3.500)	0.047*** (3.743)	0.043*** (3.365)	0.043*** (3.365)
Board			0.006 (0.412)	0.002 (0.135)	0.002 (0.135)
ROA			-0.048** (-2.210)	-0.046** (-2.154)	-0.046** (-2.154)
INST				0.080*** (4.477)	0.080*** (4.477)
Own_Conc				-0.013 (-0.428)	-0.013 (-0.428)
Constant	-0.075*** (-44.617)	-1.266*** (-9.713)	-1.271*** (-9.427)	-1.211*** (-9.105)	-1.211*** (-9.105)
Observations	30,201	30,201	30,201	30,201	30,201
R-squared	0.496	0.502	0.502	0.502	0.502
Firm FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES

Notes: Robust t-statistics in parentheses. *** p<0.01, ** p<0.05, * p<0.1

4.3. Robustness checks

To ensure the reliability of our baseline findings, we conduct a series of robustness tests. The results are presented in [Table 5](#).

We employ five robustness tests to validate our main findings. First, Models (1) and (2) use lagged independent variables (one-period and two-period lags, respectively) to mitigate potential simultaneity concerns. The coefficients on L.AI (-0.008, p < 0.05) and L2.AI (-0.007, p < 0.10) remain negative and significant, confirming that the effect of AI on green innovation bubbles persists over

Table 5
Robustness checks.

VARIABLES	(1) Lagged AI (t-1) GIB	(2) Lagged AI (t-2) GIB	(3) Alternative DV GIB_replace	(4) Excluding COVID GIB	(5) Industry+Year FE GIB
L.AI	-0.008** (-2.006)				
L2.AI		-0.007* (-1.734)			
AI			-0.006** (-2.052)	-0.011** (-2.099)	-0.010*** (-3.689)
Size	0.057*** (7.595)	0.066*** (7.631)	0.045*** (8.232)	0.039*** (4.297)	0.073*** (28.041)
Lev	-0.021 (-1.287)	-0.028 (-1.508)	-0.015 (-1.195)	-0.004 (-0.155)	-0.013 (-1.514)
ROE	0.033** (2.372)	0.021 (1.404)	0.017 (1.612)	0.038* (1.889)	0.018 (1.554)
Board	0.002 (0.147)	-0.004 (-0.225)	0.015 (1.144)	0.016 (0.715)	-0.018* (-1.898)
ROA	-0.020 (-0.877)	-0.027 (-1.069)	-0.026 (-1.392)	-0.022 (-0.749)	-0.086*** (-3.825)
INST	0.062*** (3.037)	0.055** (2.311)	0.081*** (5.199)	0.088*** (3.140)	0.023*** (3.504)
Own_Conc	-0.012 (-0.366)	0.016 (0.450)	-0.017 (-0.668)	-0.033 (-0.742)	0.034*** (2.775)
Constant	-1.364*** (-8.067)	-1.565*** (-7.940)	-1.129*** (-9.022)	-0.984*** (-4.805)	-1.672*** (-31.165)
Observations	24,531	20,874	29,541	16,259	30,201
R-squared	0.539	0.561	0.593	0.503	0.136
Firm FE	YES	YES	YES	YES	NO
Year FE	YES	YES	YES	YES	YES
Industry FE	NO	NO	NO	NO	YES

Notes: Robust t-statistics in parentheses. *** p<0.01, ** p<0.05, * p<0.1

time. Second, Model (3) employs an alternative measure of the dependent variable (GIB replace), and the coefficient on AI remains negative and significant (-0.006 , $p < 0.05$). Third, Model (4) excludes the COVID-19 pandemic period (2020-2022) to address concerns that the pandemic may have disrupted normal innovation patterns. The coefficient on AI is -0.011 ($p < 0.05$), actually larger in magnitude than the baseline estimate, suggesting our findings are not driven by pandemic-related anomalies. Fourth, Model (5) replaces firm fixed effects with industry fixed effects while retaining year fixed effects. This specification controls for time-invariant industry characteristics rather than firm-specific heterogeneity. The coefficient on AI remains negative and highly significant (-0.010 , $p < 0.01$), confirming that our results are robust to alternative model specifications. The lower R-squared (0.136) compared to the baseline model reflects the removal of firm-level absorptive capacity, yet the key finding remains unchanged.

4.4. Endogeneity tests

To address potential endogeneity concerns arising from reverse causality or omitted variables, we employ three approaches: instrumental variable (IV) estimation, propensity score matching (PSM), and Heckman two-stage correction. The results are presented in Table 6.

For the IV approach, we use lagged AI application (LAI) as an instrument for current AI application. The first-stage results in Model (1) show that the instrument is strongly correlated with the endogenous variable (coefficient = 0.183 , $p < 0.01$). The Kleibergen-Paap rk LM statistic of 184.432 ($p = 0.0000$) rejects the null hypothesis of under-identification, and the Kleibergen-Paap rk Wald F statistic of 199.150 exceeds the Stock-Yogo critical value of 16.38 , confirming instrument validity and ruling out weak instrument concerns. The second-stage results in Model (2) show that the coefficient on instrumented AI is -0.042 ($p < 0.05$), larger in magnitude than the baseline OLS estimate, suggesting that OLS may underestimate the true causal effect of AI on green innovation bubbles.

Model (3) employs propensity score matching to address selection bias. After matching firms with and without AI application based on observable characteristics, the coefficient on AI remains -0.010 ($p < 0.01$), consistent with our baseline results. Model (4) applies the Heckman two-stage correction to address potential sample selection bias. The inverse Mills ratio (IMR) is significant (-0.322 , $p < 0.01$), indicating the presence of selection bias. After correction, the coefficient on AI is -0.061 ($p < 0.01$), substantially larger than the baseline estimate, further confirming that AI application suppresses green innovation bubbles.

Collectively, these endogeneity tests provide strong evidence that the negative relationship between AI application and green innovation bubbles is causal rather than spurious, supporting Hypothesis 1.

Table 6
Endogeneity tests.

VARIABLES	(1) First Stage AI	(2) Second Stage (IV) GIB	(3) PSM GIB	(4) Heckman GIB
LAI	0.183*** (14.060)			
AI		-0.042** (-1.979)	-0.010*** (-3.022)	-0.061*** (-2.968)
IMR				-0.322*** (-2.582)
Size	0.039*** (4.681)	0.059*** (7.736)	0.050*** (8.683)	-0.031 (-0.945)
Lev	0.006 (0.196)	-0.021 (-1.270)	-0.023 (-1.641)	-0.050*** (-2.898)
ROE	0.048* (1.915)	0.035** (2.516)	0.043*** (3.407)	-0.046 (-1.258)
Board	0.031 (1.054)	0.004 (0.233)	0.002 (0.129)	0.001 (0.035)
ROA	-0.083* (-1.662)	-0.024 (-1.024)	-0.047** (-2.178)	0.223** (2.092)
INST	-0.003 (-0.075)	0.061*** (3.033)	0.079*** (4.439)	0.106*** (5.192)
Own_Conc	-0.045 (-0.787)	-0.014 (-0.426)	-0.013 (-0.428)	0.036 (1.015)
Constant	-0.662*** (-3.691)	-1.364*** (-8.07)	-1.216*** (-9.131)	1.008 (1.142)
Observations	24,531	24,351	30,190	30,201
R-squared	0.646	0.010	0.502	0.503
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Kleibergen-Paap rk LM		184.432 ($p=0.0000$)		
Kleibergen-Paap rk Wald F		199.150 (critical value=16.38)		

Notes: Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

4.5. Analysis of moderating effects

4.5.1. The strengthening role of environmental regulation

Table 7 presents results testing whether environmental regulation moderates the relationship between AI application and green innovation bubbles. Model (1) includes the main effects of AI and environmental regulation (ER), while Model (2) introduces the interaction term ($ER \times AI$) to test Hypothesis 2.

The results provide strong support for Hypothesis 2. In Model (2), the coefficient on the interaction term ($ER \times AI$) is negative and statistically significant at the 5% level (-0.028 , $t = -1.970$), indicating that environmental regulation strengthens the negative effect of AI application on green innovation bubbles. The main effect of AI remains negative and significant (-0.010 , $p < 0.01$), confirming that AI suppresses green innovation bubbles even after accounting for the moderating effect.

To interpret this interaction effect substantively, in regions with low environmental regulation, the effect of AI on GIB is primarily captured by the main coefficient (-0.010). However, in regions with stringent environmental regulation, the total effect of AI on GIB is amplified to approximately -0.038 ($-0.010 + -0.028$), representing nearly a fourfold increase in magnitude. This pattern confirms that environmental regulation serves as a critical catalyst that channels AI capabilities toward substantive innovation.

The underlying mechanism is intuitive: stringent environmental regulation creates credible threats of penalties for non-compliance, compelling firms to prioritize genuine environmental improvements over symbolic gestures. Under these conditions, firms strategically deploy their AI resources to solve real environmental challenges that can withstand regulatory scrutiny, such as emissions optimization and pollution abatement technologies. These efforts naturally generate high-quality invention patents rather than superficial utility models. The finding underscores that technology and institutions are complementary rather than substitutable forces in driving corporate sustainability.

4.5.2. The strengthening role of market competition

Table 8 presents results testing whether market competition moderates the relationship between AI application and green innovation bubbles. Model (1) includes the main effects of AI and market competition (HHI), while Model (2) introduces the interaction term ($HHI \times AI$) to test Hypothesis 3.

The results provide support for Hypothesis 3. In Model (2), the coefficient on the interaction term ($HHI \times AI$) is positive and statistically significant at the 5% level (1.178 , $t = 2.207$). Since higher HHI indicates lower competition (greater market concentration), this positive coefficient means that AI's negative effect on GIB weakens as competition concentrated, or equivalently, strengthens as competition intensifies. This finding supports H3.

To illustrate the economic significance of this moderation, we calculate the conditional effects of AI at different levels of market competition. In highly competitive industries (low HHI approaching zero), the effect of AI on GIB is captured primarily by the main coefficient (-0.009), representing a strong bubble-suppressing effect. However, in highly concentrated industries (high HHI), the

Table 7
Moderating effect of environmental regulation.

VARIABLES	(1) GIB	(2) GIB
AI	-0.010*** (-3.025)	-0.010*** (-3.140)
ER $\times$ AI		-0.028** (-1.970)
ER	0.014* (1.872)	0.013* (1.717)
Size	0.050*** (8.665)	0.050*** (8.645)
Lev	-0.022 (-1.585)	-0.022 (-1.583)
ROE	0.042*** (3.334)	0.042*** (3.315)
Board	0.002 (0.133)	0.002 (0.156)
ROA	-0.046** (-2.147)	-0.046** (-2.146)
INST	0.080*** (4.478)	0.080*** (4.480)
Own_Conc	-0.013 (-0.433)	-0.012 (-0.424)
Constant	-1.225*** (-9.155)	-1.222*** (-9.135)
Observations	30,201	30,201
R-squared	0.502	0.502
Firm FE	YES	YES
Year FE	YES	YES

Notes: Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 8
Moderating effect of market competition.

VARIABLES	(1) GIB	(2) GIB
AI	-0.010*** (-3.011)	-0.009*** (-2.758)
HHI × AI		1.178** (2.207)
HHI	0.102* (1.715)	0.406** (2.512)
Size	0.050*** (8.664)	0.050*** (8.675)
Lev	-0.023 (-1.604)	-0.022 (-1.547)
ROE	0.043*** (3.362)	0.043*** (3.373)
Board	0.002 (0.123)	0.002 (0.127)
ROA	-0.047** (-2.166)	-0.047** (-2.180)
INST	0.080*** (4.478)	0.077*** (4.349)
Own_Conc	-0.013 (-0.454)	-0.012 (-0.407)
Constant	-1.213*** (-9.116)	-1.216*** (-9.136)
Observations	30,201	30,201
R-squared	0.502	0.503
Firm FE	YES	YES
Year FE	YES	YES

Notes: Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

positive interaction term substantially offsets the main effect. For instance, when HHI exceeds approximately 0.0076 (roughly 0.4 standard deviations above the mean), the total effect of AI on GIB becomes positive, suggesting that AI may actually facilitate green innovation bubbles in highly concentrated markets.

The underlying mechanism reflects competition's disciplining role on corporate resource allocation. In competitive markets, firms face intense scrutiny from customers, investors, and competitors, raising the costs and risks of superficial greenwashing. This external pressure compels firms to deploy AI resources toward substantive innovations that generate genuine competitive advantages rather than cosmetic green credentials. Conversely, firms in concentrated markets enjoy greater market power and weaker external discipline, reducing the urgency to maximize AI's substantive innovation potential. The finding underscores that market structure critically shapes how firms deploy transformative technologies, with competition serving as a governance mechanism that channels AI toward genuine environmental progress.

4.5.3. Comparative analysis of moderating effects

Fig. 1 visualizes the two moderating effects. Panel A illustrates the environmental regulation moderation, while Panel B depicts the market competition moderation.

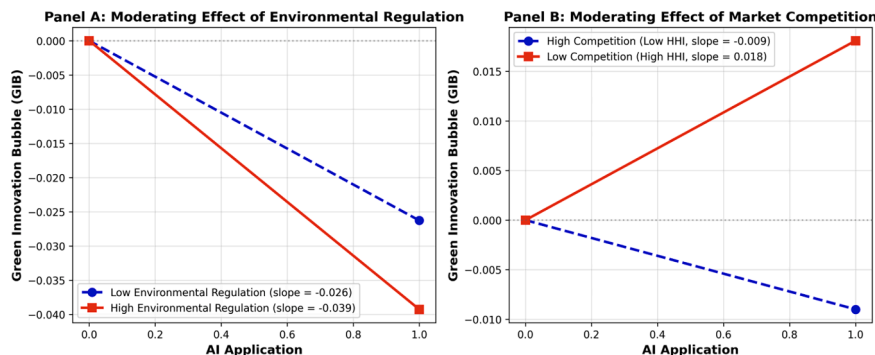


Fig. 1. Slope plots of moderating effects.

Notes: Panel A shows the conditional effect of AI application on green innovation bubbles at high (mean + 1 S.D.) and low (mean - 1 S.D.) levels of environmental regulation. Panel B shows the conditional effect at high competition (low HHI) and low competition (high HHI) levels. Slopes are calculated based on regression coefficients from Tables 7 and 8.

In Panel A, both slopes are negative, but the slope under high environmental regulation (solid line) is steeper than under low regulation (dashed line). This confirms that stringent environmental regulation amplifies AI's bubble-suppressing effect. The difference in slopes (-0.039 vs. -0.026) indicates that AI's effectiveness increases by approximately 50% when environmental regulation is high.

In Panel B, the two slopes diverge dramatically. Under high competition (low HHI, dashed line), the slope is negative (-0.009), indicating that AI effectively suppresses green innovation bubbles when firms face intense competitive pressure. However, under low competition (high HHI, solid line), the slope becomes positive (+0.018), suggesting that AI may actually facilitate green innovation bubbles in concentrated markets where competitive discipline is absent. This reversal effect powerfully demonstrates that market competition serves as a critical governance mechanism that determines whether AI functions as a "substance driver" or a "fiction booster."

Comparing the two moderators, both environmental regulation and market competition amplify AI's bubble-suppressing effect, though through different mechanisms institutional coercion versus market discipline. Notably, the market competition moderation ($\beta = 1.178$) is substantially larger in absolute magnitude than the environmental regulation moderation ($\beta = -0.028$), indicating that competitive pressure exerts a more pronounced influence on how firms deploy AI technologies. These findings suggest that policy-makers should strengthen both environmental regulation and competitive market structures to maximize AI's contribution to genuine green innovation. In particular, attention should be paid to preventing excessive market concentration, which may undermine AI's positive environmental effects.

4.6. Heterogeneity analysis

To explore the boundary conditions of AI's impact on green innovation bubbles, we conduct heterogeneity analyses across three dimensions: ownership structure, pollution intensity, and location in AI innovation pilot zones. The results are presented in Table 9.

4.6.1. Heterogeneity by state ownership

Panel A reveals a striking divergence between SOEs and non-SOEs. For non-SOEs, AI application exhibits a strong negative effect on GIB ($\beta = -0.025$, $p < 0.01$), while for SOEs, the coefficient is positive and insignificant ($\beta = 0.014$, $p > 0.1$). This suggests that AI's bubble-suppressing effect is concentrated entirely in non-SOEs. The underlying mechanism reflects differences in governance structures and incentive systems: non-SOEs face harder budget constraints and stronger external monitoring from private investors, compelling them to deploy AI resources efficiently toward substantive innovation. Conversely, SOEs' softer incentive structures and political mandates may tolerate or even encourage visible "green credentials" regardless of quality, neutralizing AI's potential benefits.

4.6.2. Heterogeneity by pollution intensity

Panel B shows a similar pattern across pollution intensity. For non-heavy-polluting firms, AI significantly suppresses GIB ($\beta = -0.016$, $p < 0.01$), whereas for heavy-polluting firms, the effect is positive and insignificant ($\beta = 0.014$, $p > 0.1$). This counterintuitive finding suggests that heavy-polluting firms, despite facing greater regulatory scrutiny, may face stronger legitimacy pressures that incentivize quantity-focused green patenting strategies. These firms may deploy AI to generate more visible green credentials rather than to solve genuine environmental challenges. In contrast, non-heavy-polluting firms face less acute legitimacy threats, allowing AI to be channeled toward quality-oriented innovation.

4.6.3. Heterogeneity by AI pilot zone

Panel C indicates that firms located in AI innovation pilot zones exhibit a stronger bubble-suppressing effect ($\beta = -0.017$, $p < 0.01$) compared to those outside pilot zones ($\beta = -0.009$, $p < 0.1$). Both coefficients are negative and significant, but the magnitude is nearly double for pilot zone firms. This suggests that superior AI infrastructure, concentrated talent pools, and policy support in pilot zones enhance firms' capacity to leverage AI for substantive green innovation. The ecosystem advantages facilitate deeper technological integration and accelerate organizational learning, enabling more effective deployment of AI toward genuine R&D rather than superficial applications.

These heterogeneity findings underscore that AI's role in suppressing green innovation bubbles is neither universal nor unconditional. Ownership structure, pollution intensity, and regional policy environment jointly shape how firms deploy AI technologies, highlighting the importance of contextual factors in understanding technology-sustainability relationships.

5. Discussion and conclusion

5.1. Main findings

This study investigates whether AI serves as a "substance driver" that promotes genuine green innovation or a "fiction booster" that facilitates sophisticated greenwashing. Analyzing 30,201 firm-year observations from Chinese A-share listed manufacturing companies (2013 to 2023), we obtain the following key findings.

First, the baseline results demonstrate that AI application significantly suppresses green innovation bubbles ($\beta = -0.010$, $p < 0.01$), supporting Hypothesis 1. This effect remains robust across alternative specifications, lagged variables, instrumental variable estimation, PSM, and Heckman correction, confirming a causal relationship.

Second, the moderating analyses reveal that AI's impact is highly contingent on external contexts. Environmental regulation amplifies AI's bubble-suppressing effect ($ER \times AI = -0.028$, $p < 0.05$), supporting Hypothesis 2. Additionally, market competition

Table 9
Heterogeneity analysis.

Panel A: Heterogeneity by State Ownership		
VARIABLES	SOEs GIB	Non-SOEs GIB
AI	0.014 (1.628)	-0.025*** (-8.681)
Size	0.044*** (3.555)	0.058*** (8.229)
Lev	-0.019 (-0.523)	-0.009 (-0.633)
ROE	0.029 (1.114)	0.026* (1.820)
Board	0.002 (0.050)	0.007 (0.513)
ROA	-0.082* (-1.873)	-0.017 (-0.703)
INST	0.158*** (3.506)	0.062*** (3.072)
Own_Conc	-0.126** (-2.014)	0.007 (0.192)
Constant	-1.084*** (-3.846)	-1.400*** (-8.649)
Observations	9,911	20,270
R-squared	0.543	0.460
Firm FE	YES	YES
Year FE	YES	YES
Panel B: Heterogeneity by Pollution Intensity		
VARIABLES	Heavy-Polluting GIB	Non-Heavy-Polluting GIB
AI	0.014 (1.474)	-0.016*** (-4.472)
Size	0.044*** (4.011)	0.055*** (7.745)
Lev	-0.051* (-1.760)	-0.017 (-1.043)
ROE	0.058** (2.223)	0.040*** (2.725)
Board	0.020 (0.683)	-0.009 (-0.538)
ROA	-0.144*** (-2.738)	-0.022 (-0.928)
INST	0.048 (1.338)	0.092*** (4.395)
Own_Conc	0.136** (2.212)	-0.048 (-1.408)
Constant	-1.139*** (-4.656)	-1.282*** (-7.802)
Observations	6,341	23,860
R-squared	0.474	0.516
Firm FE	YES	YES
Year FE	YES	YES
Panel C: Heterogeneity by AI Pilot Zone		
VARIABLES	Pilot Zone GIB	Non-Pilot Zone GIB
AI	-0.017*** (-3.343)	-0.009* (-1.925)
Size	0.052*** (4.277)	0.042*** (6.092)
Lev	0.017 (0.619)	-0.008 (-0.385)
ROE	0.037* (1.753)	0.038** (2.252)
Board	-0.006 (-0.193)	0.008 (0.454)
ROA	-0.036 (-0.928)	-0.035 (-1.314)
INST	-0.029	0.077***

(continued on next page)

Table 9 (continued)

Panel C: Heterogeneity by AI Pilot Zone		
VARIABLES	Pilot Zone GIB	Non-Pilot Zone GIB
	(-0.824)	(3.148)
Own_Conc	-0.093 (-1.144)	0.022 (0.578)
Constant	-1.180*** (-4.443)	-1.053*** (-6.820)
Observations	11,725	18,347
R-squared	0.741	0.470
Firm FE	YES	YES
Year FE	YES	YES

Notes: Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

strengthens this effect ($\text{HHI} \times \text{AI} = 1.178$, $p < 0.05$), with competitive pressure serving as a disciplining mechanism that channels AI toward substantive innovation, supporting Hypothesis 3. In contrast, market concentration weakens AI's bubble-suppressing effect, potentially even reversing it in highly concentrated industries.

Third, heterogeneity analyses uncover important boundary conditions. AI's effectiveness is concentrated in non-SOEs ($\beta = -0.025$, $p < 0.01$) and non-heavy-polluting firms ($\beta = -0.016$, $p < 0.01$), while SOEs and heavy-polluting firms show insignificant effects. Firms in AI pilot zones exhibit stronger effects ($\beta = -0.017$) than those outside ($\beta = -0.009$), though both remain significant.

5.2. Theoretical contributions

This study makes three primary contributions. First, we reveal AI's conditional nature in corporate sustainability. Rather than viewing AI as universally beneficial, our findings demonstrate that AI can facilitate either substantive progress or sophisticated greenwashing depending on institutional and market contexts. This challenges technologically deterministic narratives prevalent in existing literature.

Second, we develop an integrated "institution-market" dual contingency framework. Unlike prior research examining regulation or competition in isolation, we simultaneously consider these distinct yet complementary forces. Our framework demonstrates that both institutional pressure (environmental regulation) and market discipline (competition) channel AI toward substantive green innovation, though through different mechanisms.

Third, we uncover counterintuitive heterogeneity patterns. The finding that AI's benefits concentrate in non-SOEs and non-heavy-polluting firms, rather than where environmental pressures are most acute, suggests that governance efficiency and legitimacy pressures critically mediate technology effects.

5.3. Practical implications

For corporate managers, simply investing in AI is insufficient. Strategic deployment matters critically, particularly resisting short-term pressures to allocate AI toward marketing rather than substantive R&D. Non-SOEs should leverage their governance advantages to maximize AI's innovation benefits.

For policymakers, technology policy and environmental policy must be designed synergistically. Strengthening environmental regulation enhances AI's positive effects. Equally important, maintaining competitive market structures ensures that AI is deployed for substantive innovation rather than superficial greenwashing. Policymakers should be vigilant against excessive market concentration, which may undermine AI's positive environmental effects.

5.4. Limitations and future research

This study has several limitations. First, our GIB measure, while objective, remains a proxy for innovation quality. Future research could employ patent citation analysis or textual analysis for more granular quality assessment. Second, our AI measure captures investment but not specific deployment strategies. Case studies could illuminate internal mechanisms more deeply. Third, the insignificant AI effects in SOEs and heavy-polluting firms warrant further investigation. Whether this reflects measurement issues or genuine institutional constraints remains unclear. Finally, future research could explore the dynamic evolution of AI's impact over time, testing whether a "J-curve" effect exists as firms develop complementary capabilities.

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CRedit authorship contribution statement

Kai Zhao: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiaoxi Liu:** Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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